

COGS 45 Week 2 Lab Assignment

COGS 45 Week 2.

Perception and Representational Alignment Homework

1. In class, we used the data from Peterson et al. 2018 to compare the similarity of network activations in AlexNet to human similarity judgments for animals. Here, do the same thing, but for the fruits and vegetables datasets. Code and instructions are provided in the accompanying .ipynb notebooks download and upload into colab).

(A). Plot the measure of alignment (either r or r^2 are fine) for fruits and animals for different network layers ([layer names: "features[1]", "features[4]", "features[7]", "features[9]", "classifier[2]", "classifier[5]"] commands to do so are provided in homework code. Feel free to work together!).

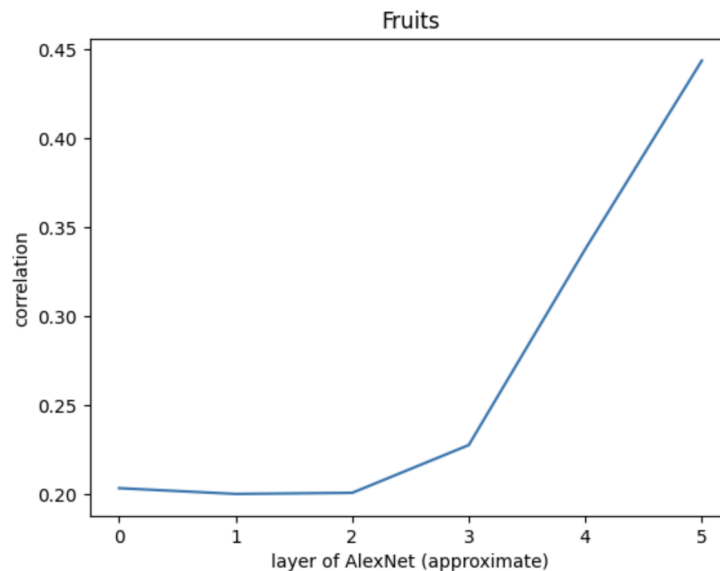


Figure 1

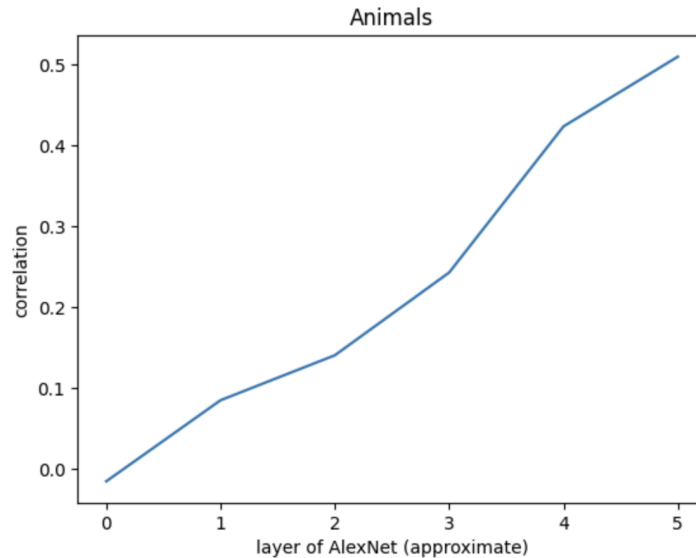


Figure 2

(B). In a sentence or two, describe your results qualitatively both across layers and across the two categories.

Layers for fruits and animals both show an increase in alignment, as we go deeper in the network. During the early layer phase, networks process images through edges and colors. During the middle layer phase, they see an increase because they move from pixels to big-picture perception, which is similar to how humans see. Closer to final layers, humans and networks have a strong correlation because at this stage, fruits and animals are perceived as objects and categories.

Animals start from 0, in comparison to fruits at 0.20. However, animals increase gradually while fruits remain flat until they reach layer two. For animals, they reach highest at 0.5 while animals see a jump in layer four. This indicates that the more layer neural networks have, the more similar they're to how humans perceive fruit and animal features.

Human representations for animals need higher-level semantic structure (where the eyes are located, a giraffe is structured differently from a chicken), whereas similarity for fruits depend on low-level visual features (a tomato and an apple are alike).

2. Here, we are simply introducing tools for measuring representational alignment between humans and conv nets. However, it is often assumed that in many cases AI increasing “representational alignment” between humans and AI systems is desirable.

Suggest:

(2a) one case in which representational alignment is, in fact, good and should be pursued. Give a brief explanation for why and how you would begin to think about accomplishing it (i.e., what measurements would you specifically want to align?).

Representational alignment is important for self-driving cars because we want them to interpret and perceive the world the same way we do for safety concerns. Both humans and the model should represent signs like stop, pedestrian walking, in a similar way. Object categorization would be desirable in this case, for example, if a deer and a human are passing by, they should be able to categorize the two and choose to save human life.

(2b) Suggest one case in which representational dis-alignment (or representational diversity) between humans and machines could actually be desirable.

Representational dis-alignment can be desirable for situations where important decisions are affected by systematic biases. For example, Black people have been historically accused of crime in Chicago due to police racism. In these situations, AI systems can detect patterns we often miss. It's similar to the idea of Lateral Thinking, in which AI provides us alternative options to consider in addition to our own human reasoning.

Bonus 1. Evaluate how the network's ability to categorize objects is/isn't affected by the human alignment fine-tuning process we used in class (i.e., minimizing dissimilarity with human representational similarity by changing weights in the fully-connected layers).

Bonus 2. Pick a second pre-trained network (e.g., ResNet, VGG — both available in torchvision) and compare its alignment with human judgments to AlexNet's. Does a deeper or more modern architecture produce more human-like representations?